RCS02 | Student Packet

CTS Cohort 6 | CTS Day 10 | 2026-07-21 | 10:00 AM-3:00 PM | 6 students

Residential Framing Basics: Measure, Layout, Build

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Big idea: Accurate framing begins before the saw starts: read the plan, measure twice, mark clearly, and verify the cut list.

Learning targets

- Read a tape measure to 1/16 inch and add common fractions.
- Identify plates, studs, king studs, jack studs, header, sill, and cripples.
- Lay out a 4-foot by 3-foot training wall with a framed window opening.
- Complete an instructor-approved cut list and assemble a square wall frame.

Today's run

10:00-10:15 | Arrival, PPE, retrieval warm-up

10:15-10:30 | Morning quiz

10:30-10:50 | Quiz review and measurement misconception check

10:50-11:20 | Tape-measure and fraction model

11:20-12:00 | Wall-member map and window-framing demonstration

12:00-12:30 | Cut-list guided practice and lab release

12:30-1:00 | Lunch and controlled reset

1:00-2:25 | Individual capstone framing lab

2:25-2:45 | Dimensions, diagonal QC, evidence, cleanup

2:45-3:00 | Exit ticket and Day 3 preview

Key words

top plate | bottom plate | stud | king stud | jack stud | header | sill | cripple | on-center | crown | square

Guided worksheet

1. Convert: 1/2 = __/16; 3/4 = __/16; 7/8 = __/16.

_____

_____

2. Add: 14 3/8 + 9 7/16. Show the denominator step.

_____

_____

3. Label the wall members on the instructor-provided drawing.

_____

_____

4. Write the cut-list quantity, length, and purpose for each member.

_____

_____

5. Record two diagonal measurements and explain what they show about square.

_____

_____

Afternoon lab mission

Each student begins an individual 4-foot by 3-foot wall capstone with a framed window opening. Students inventory stock, complete the approved cut list, mark and verify all members, make supervised cuts only after clearance, dry-fit, fasten, and complete a square/dimension inspection.

Safety boundary

- Eye and hearing protection in active cutting zones
- Instructor authorization before each cutting operation
- Guards stay installed and functional
- Clamp or support stock
- Wall stays flat until instructor authorizes movement
- Stop work for conflicting measurements or damaged material

Build / test sequence

[ ] PPE and station inspection

[ ] Read training drawing

[ ] Create cut list

[ ] Mark and partner-verify

[ ] Instructor cut authorization

[ ] Supervised cutting

[ ] Dry layout and member labels

[ ] Fasten frame

[ ] Measure diagonals and correct

[ ] Photograph, document, reset

Evidence record

[ ] Completed measurement sheet: _____

[ ] Approved cut list: _____

[ ] Labeled dry layout photo: _____

[ ] Final wall dimensions and diagonal check: _____

[ ] Instructor tool-control observation: _____

Reflection and exit ticket

One measurement skill I can now demonstrate

_____

_____

One wall member and its job

_____

_____

One verification I completed

_____

_____

One step I need to practice

_____

_____

Instructor verification

Status: [ ] Approved [ ] Correct and recheck [ ] Needs follow-up

Instructor initials: _____ Date/time: _____

RCS02 | Morning Quiz

CTS Cohort 6 | CTS Day 10 | 2026-07-21 | 10:00 AM-3:00 PM | 6 students

Residential Framing Basics: Measure, Layout, Build

Name: _____ Date: _____

Answer from your own understanding. This is a teaching check, not a trap.

1. What is the smallest graduation students must read today?

_____

_____

2. What member runs horizontally at the bottom of a framed wall?

_____

_____

3. What stud directly supports a header?

_____

_____

4. What stud runs full height beside the jack stud?

_____

_____

5. Why measure diagonals?

_____

_____

6. What must happen before any cut?

_____

_____

7. What does on-center describe?

_____

_____

8. Name one reason lumber is inspected before use.

_____

_____

9. What is the day's core professional habit?

_____

_____

10. When may a student bypass a tool guard?

_____

_____